Sri Lanka Institute of Information Technology

IT3010 – Network Design and Management

Lab 02 – Report

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2.1 (WE)

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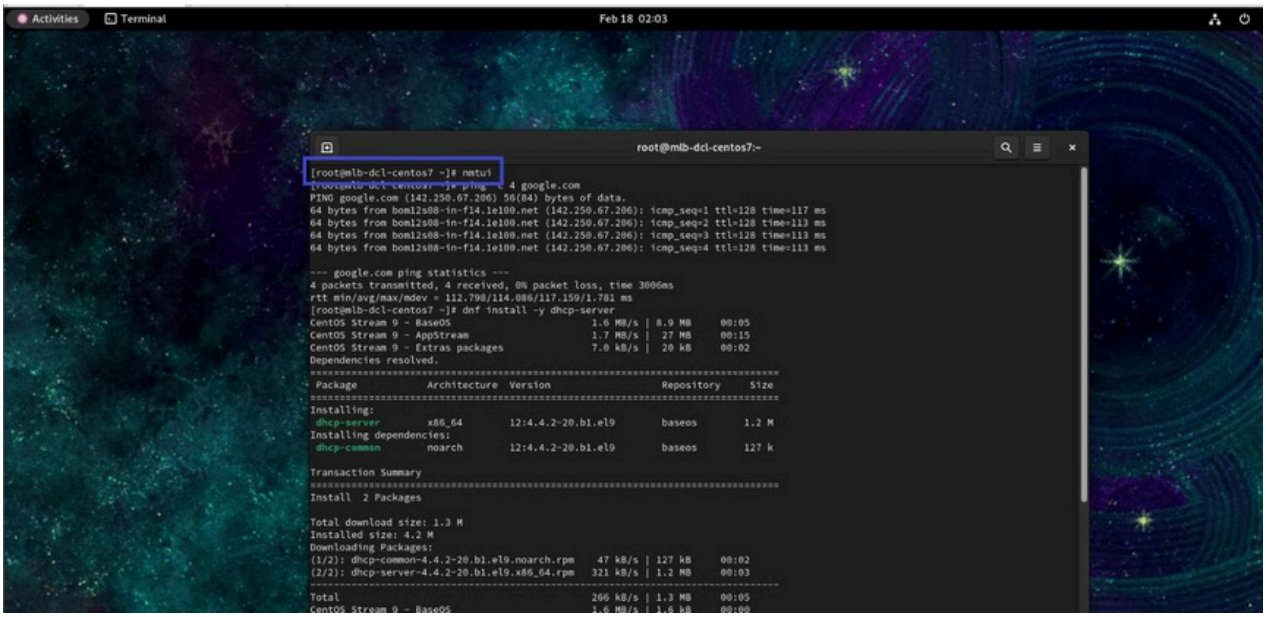
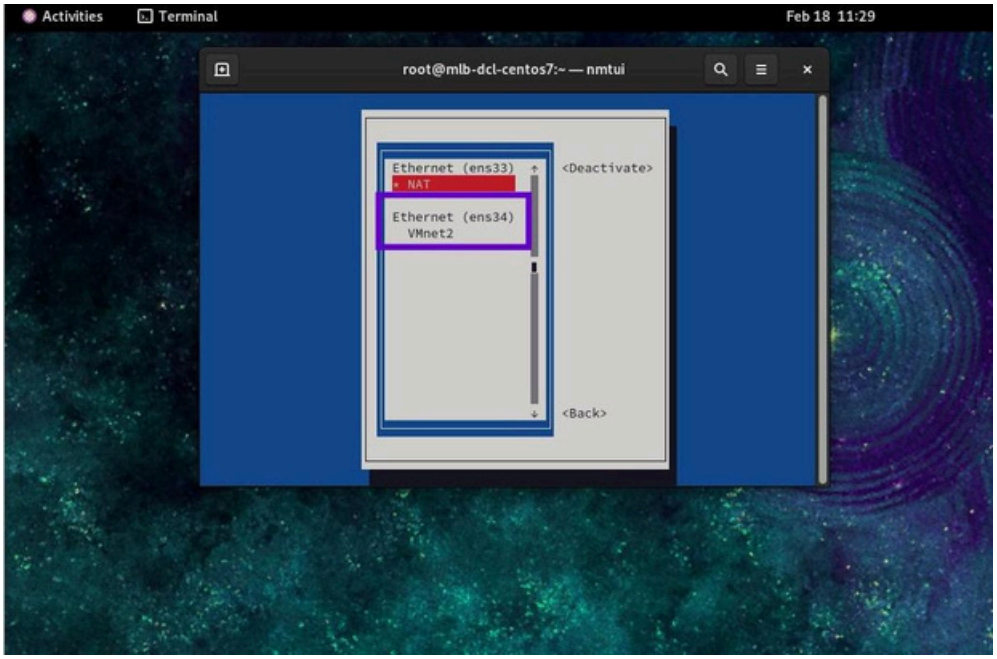
Server

Checkthe connection



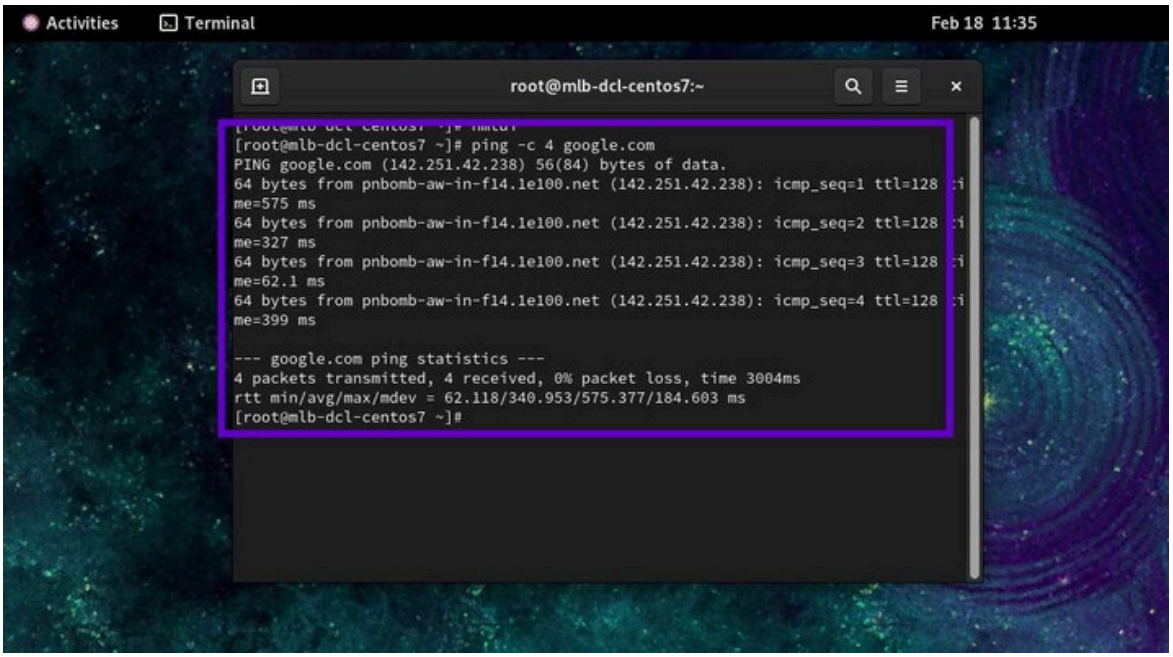
In Terminal type **nmtui**

Deactivate the **VMnet2** and **Activate NAT**



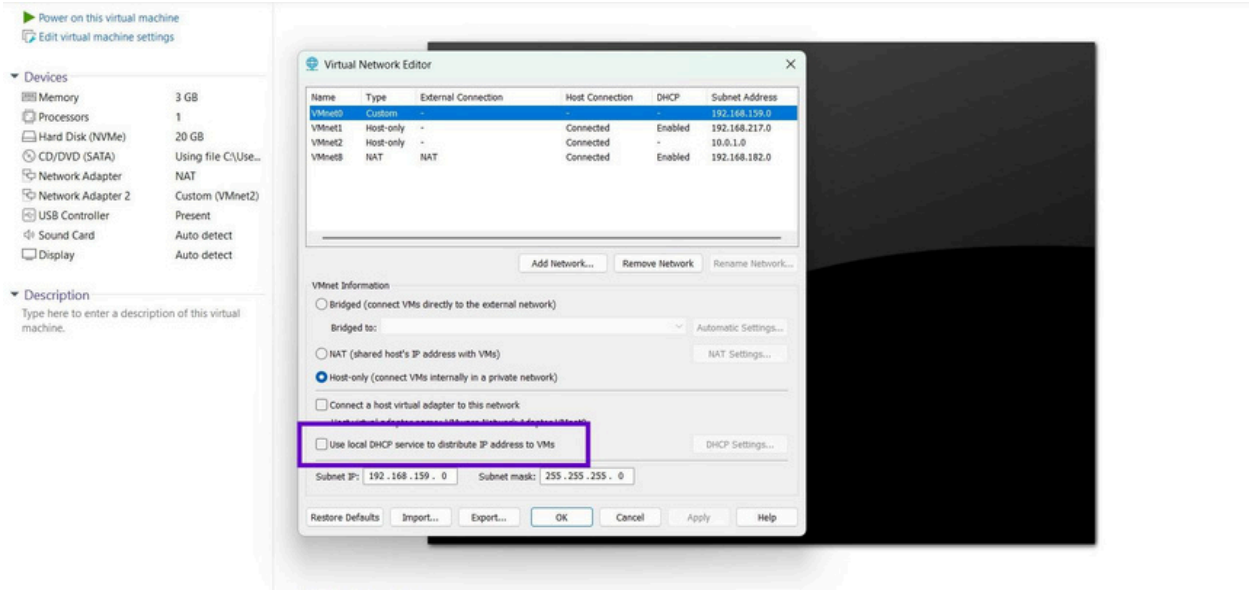
Check internet connection with ping command

ping -c 4 google.com



Step 01

Disable the DHCP settings in VMnet 2.



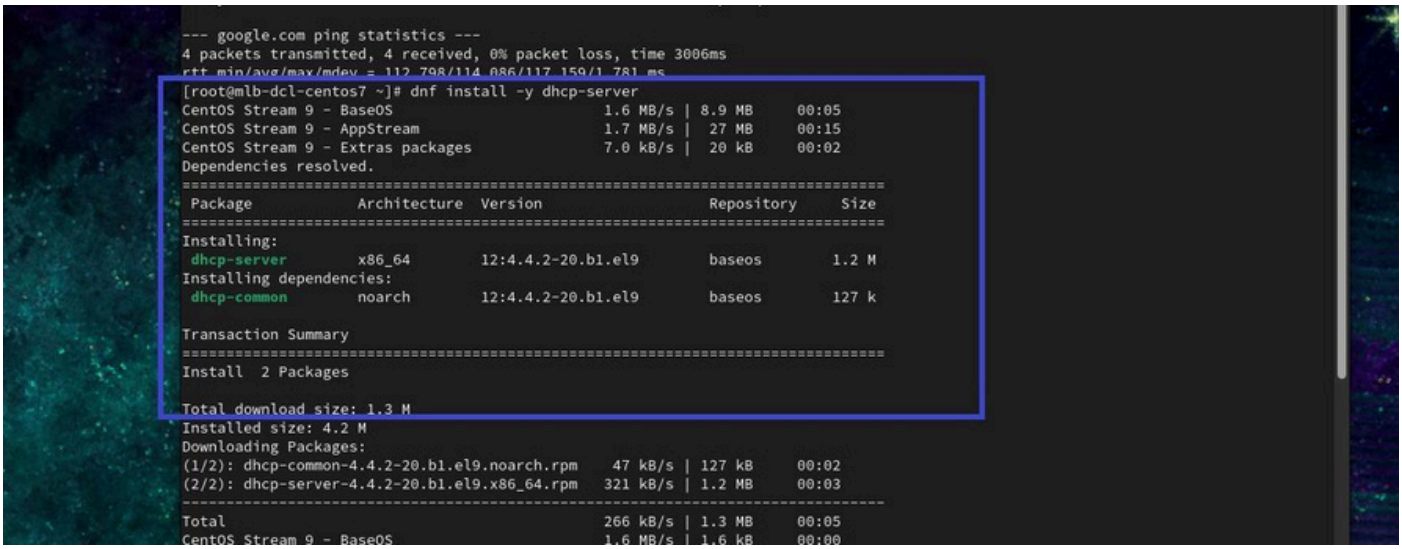
- First go to **Edit** tab in Vmware Workstation window.
- Then click select **Virtual Network editor**.
- In that pop up window has **change setting** button then click it
- Select **VMnet2**
- **Untick DHCP service** in the VMnet information section

Step 02

Installing DHCP in Centos.

ToinstallDHCPserveronCentOS, enter the following command

dnf install -y dhcp-server



Step 03

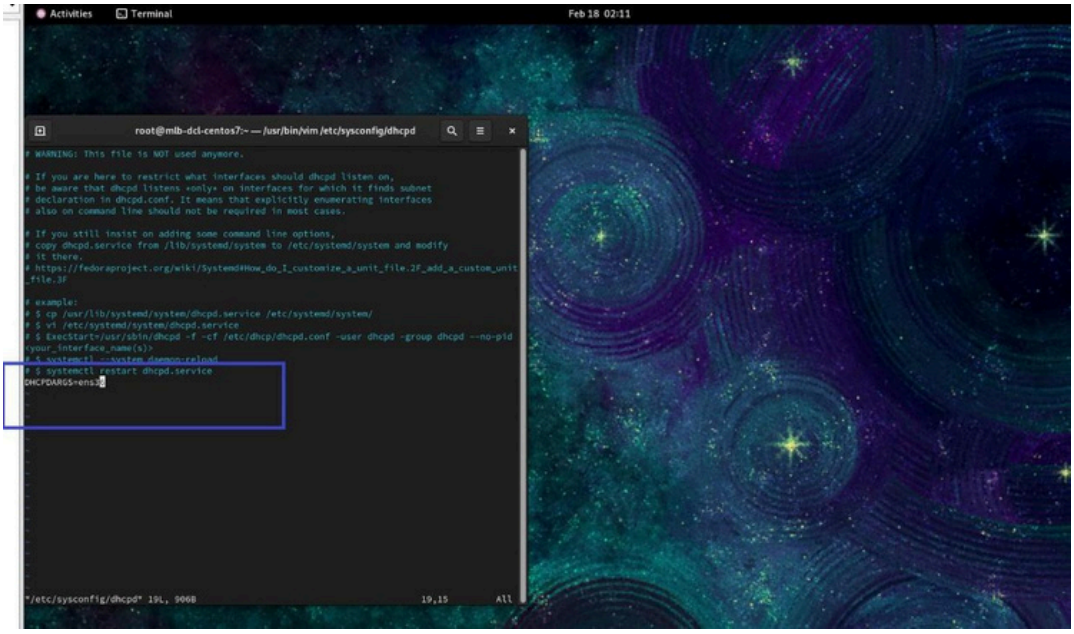
Configuring DHCP settings.

- Todothat,editfile/etc/sysconfig/dhcpd

vi /etc/sysconfig/dhcpd

- after open that directory press i to change to the insert mode
- in add the line DHCPDARGS to do that

DHCPDARGS=ens34



- After inserting, press Esc button
- After that type :wq! (Save the file)

- Copy the sample dhcp configuration file to /etc/dhcp/ directory

cp /usr/share/doc/dhcp-4.1.1/dhcpd.conf.sample /etc/dhcp/dhcpd.conf

- After this command terminal will asked a question the type **yes**

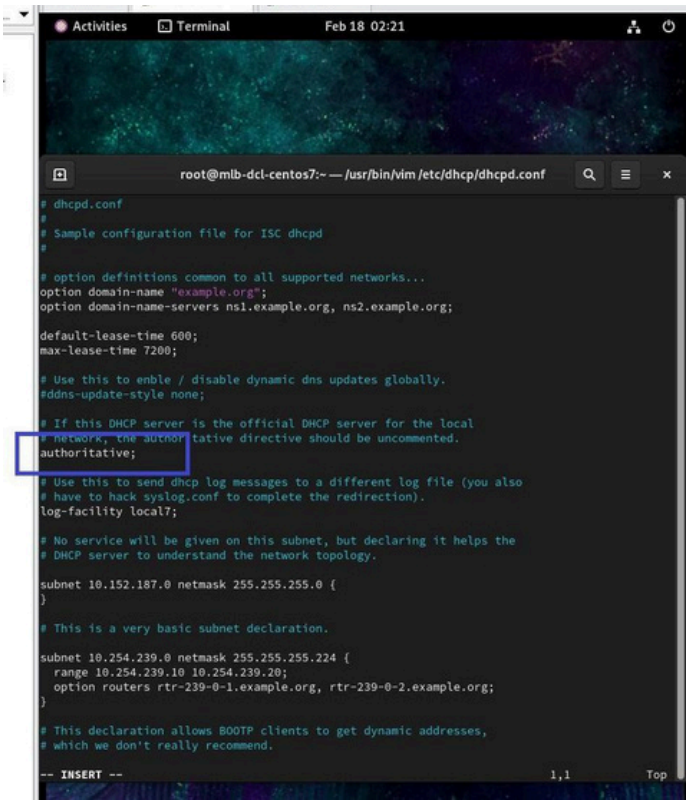
```
root@mlb-dc1-centos7 ~]# cp /usr/share/doc/dhcp-4.2.5/dhcpd.conf.example /etc/dhcp/dhcpd.conf
p: overwrite '/etc/dhcp/dhcpd.conf'? yes
```

- Now, edit dhcpd.conf file,

vi /etc/dhcp/dhcpd.conf

  - Press i
  - IfthisDHCPserver is the official DHCP server for the local network, you should uncomment the following line:

[...] authoritative; [...]



- Define the sunbet, range of ip addresses, domain and domain name servers like below:

  [...] # A slightly different configuration for an internal subnet.

  subnet 10.0.1.0 netmask 255.255.255.0 {

  range 10.0.1.20 10.0.1.30;

  # option domain-name-servers server.unixmen.local;

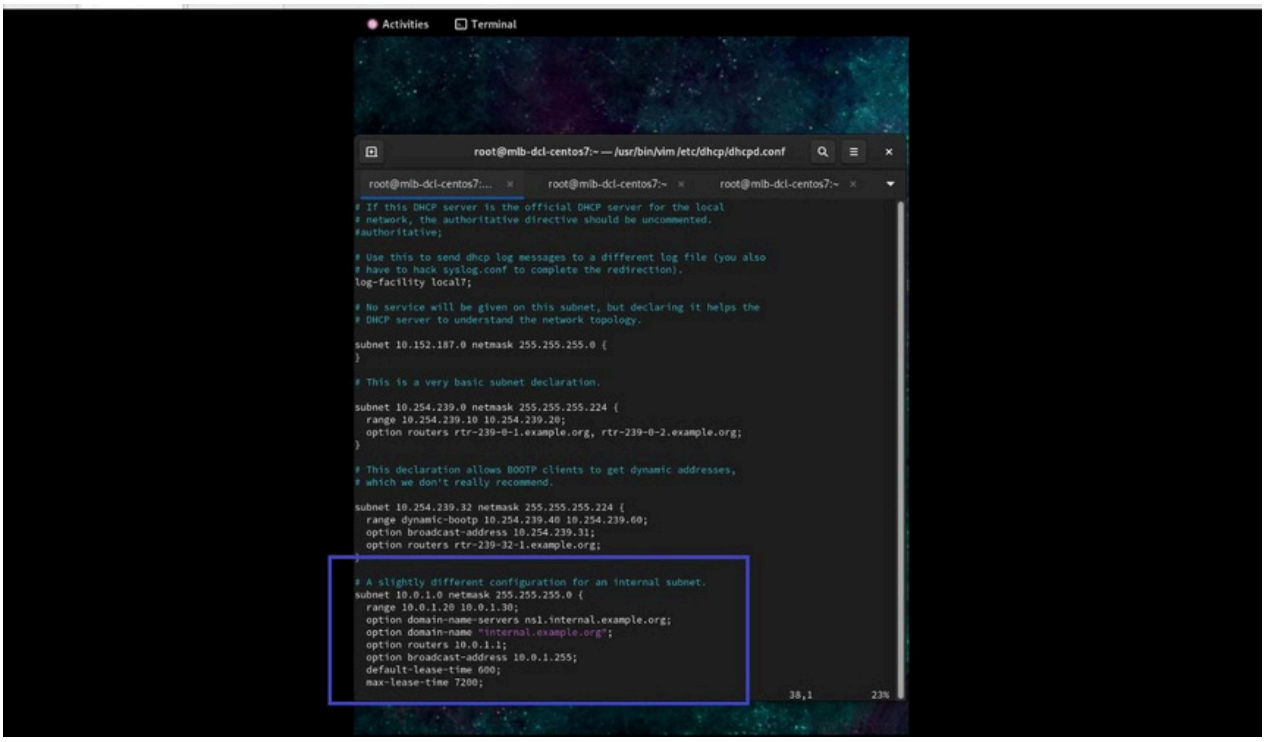
  2 option domain-name "dsnm.sub";

  option routers 10.0.1.1;

  option broadcast-address 10.0.1.255;

  default-lease-time 600;

  max-lease-time 7200;



- After all press **Esc** and type **:wq!**

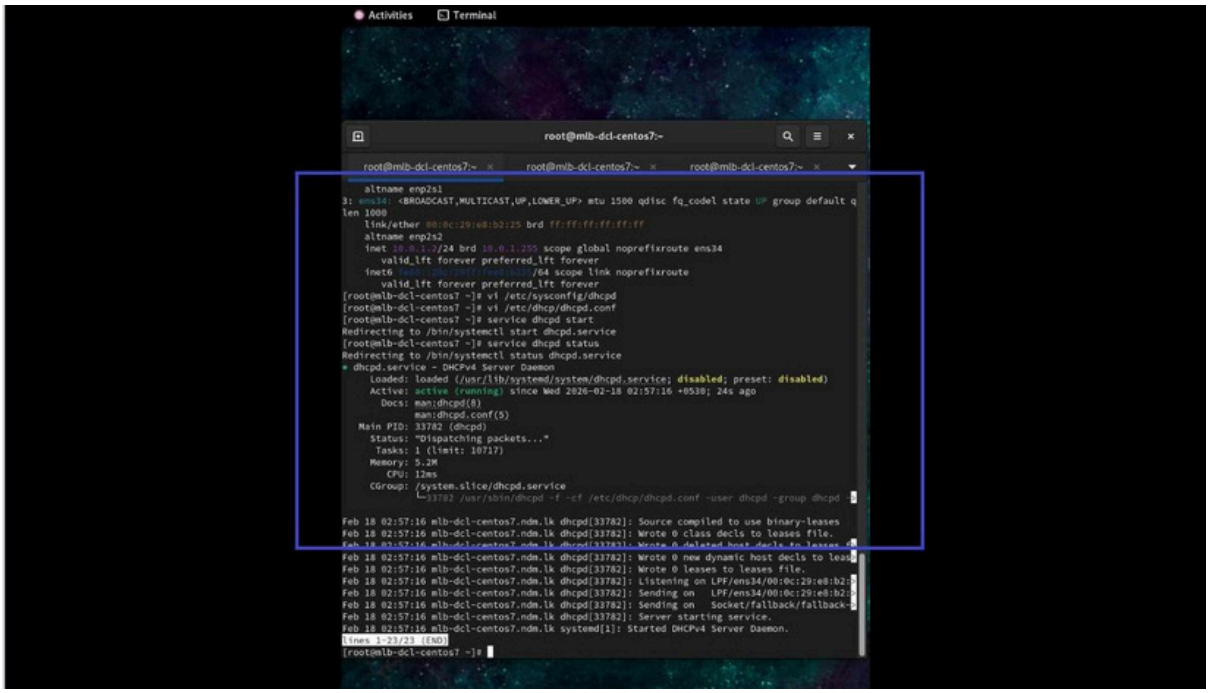
- start the dhcpd service

  service dhcpd start

Step 04

Checking the status of DHCP server and client.

Systemctl status dhcpd



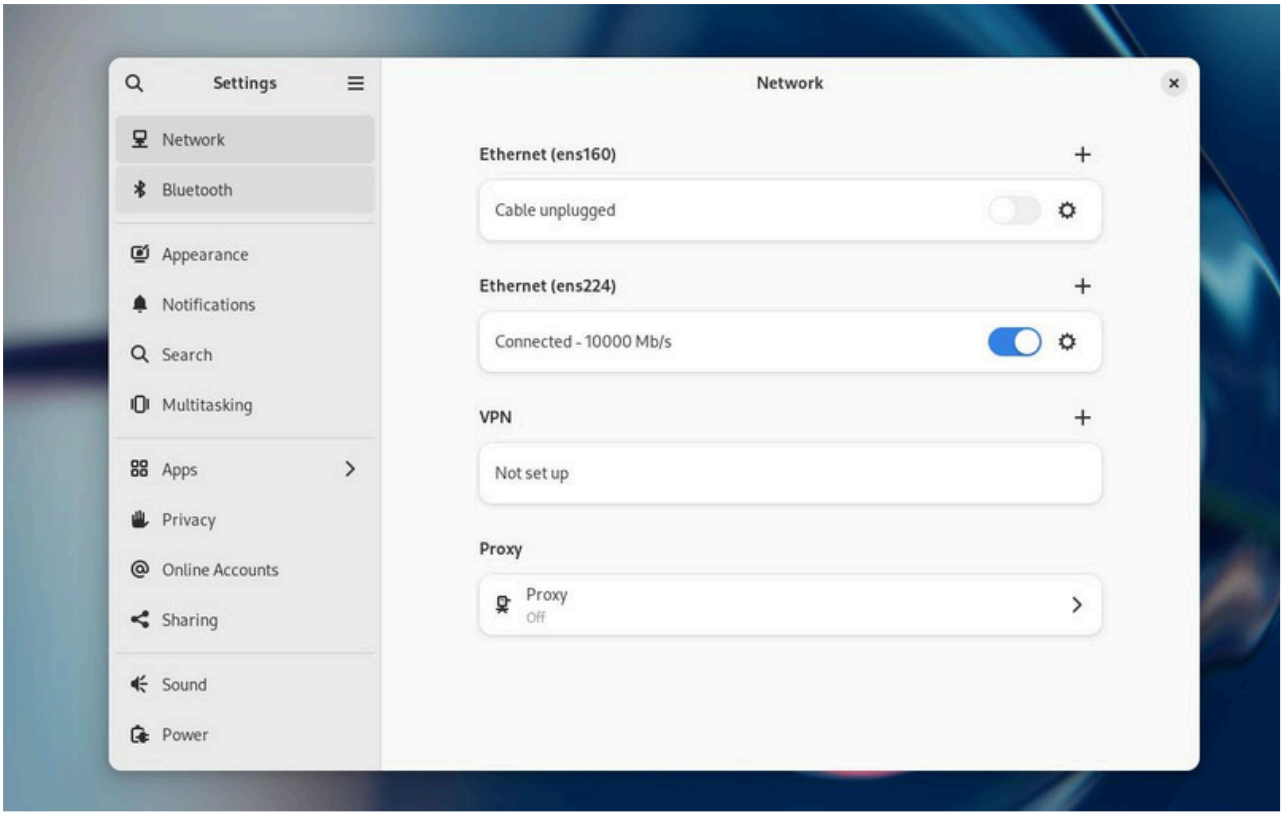
- DHCP must active and running now.

Client

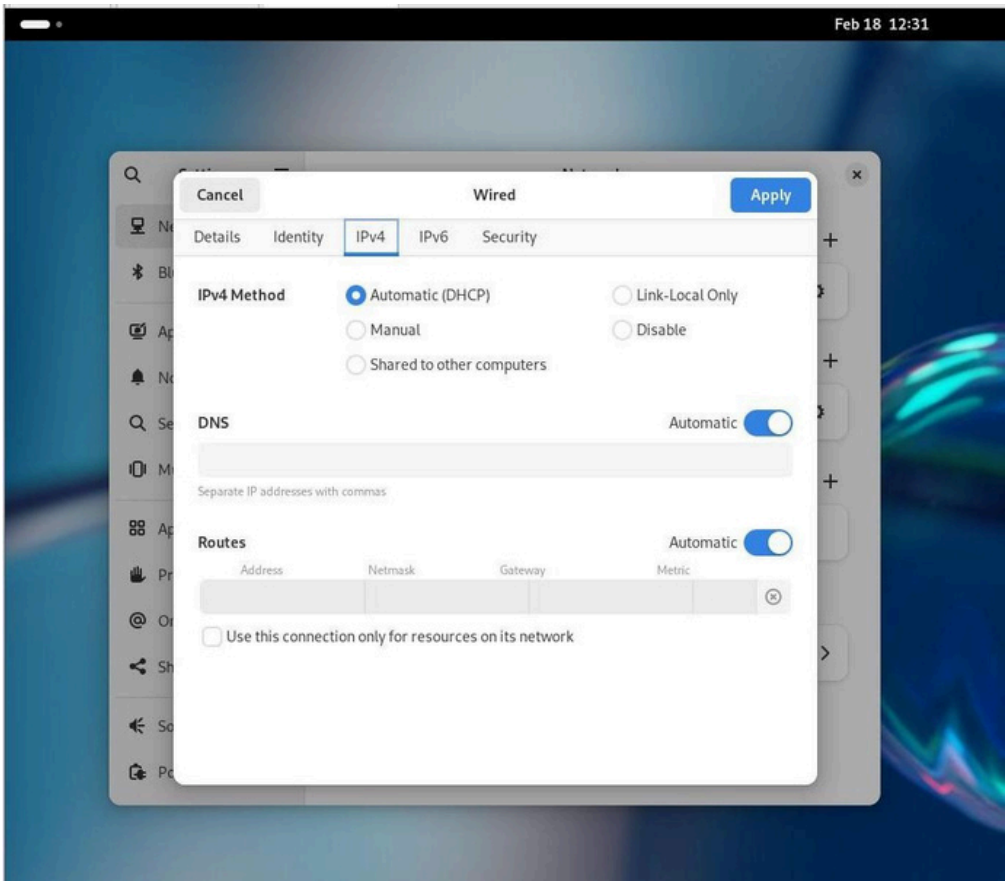
Changethe IP settings to Automatic (DHCP)

gototheclient’snetworkconfigrationsettingsandchangetheIP settings to Automatic (DHCP).

- Select VMnet2



- Change IPv4 Method to Automatic

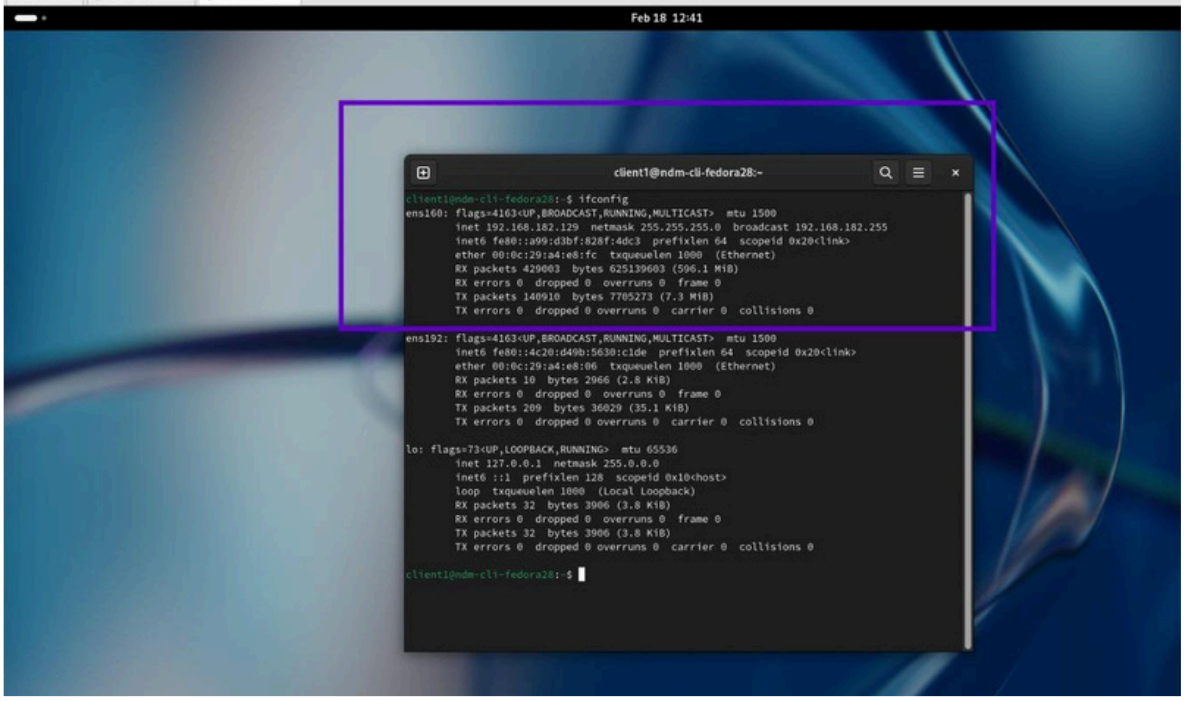


Self-Study

AssigningaTemporary IP Address

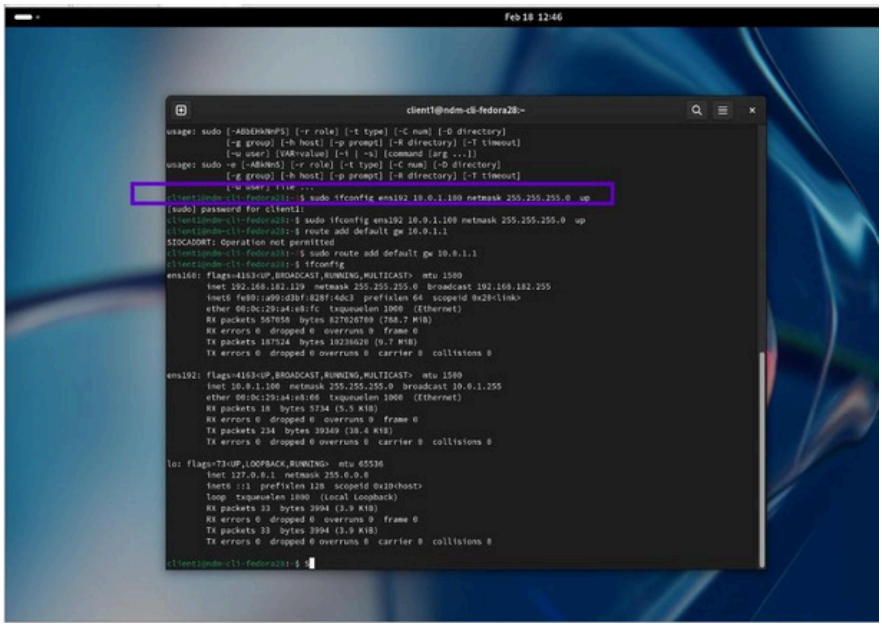
1. Identifytheinterfaceyou needto configureusing the **ifconfig** command.

2. Select VMnet2 interface (ens192)



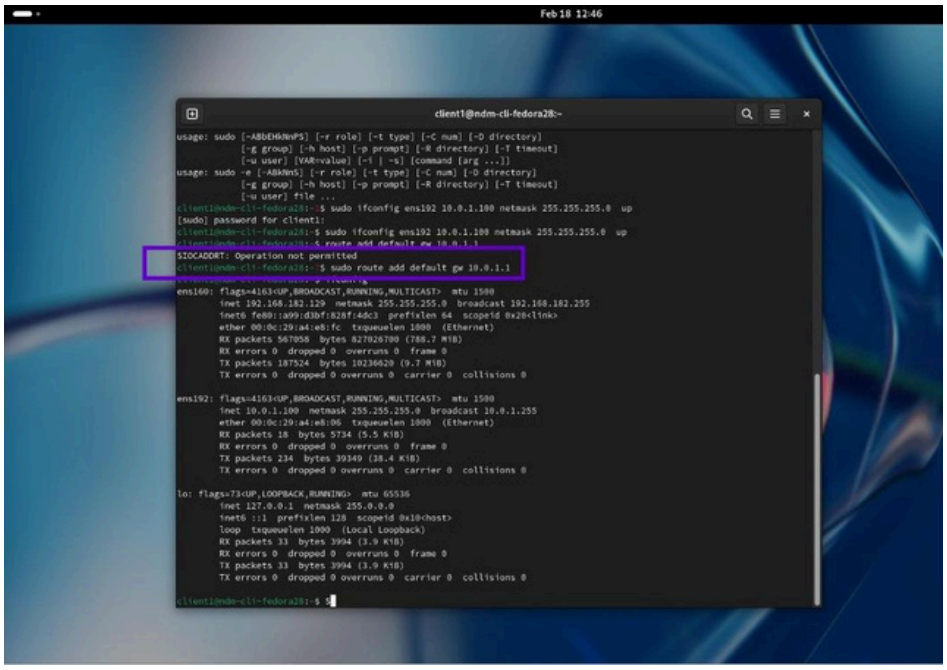
3. Assign the temporary IP;

ifconfig <interface name> up <IP Address> netmask <subnetmask>



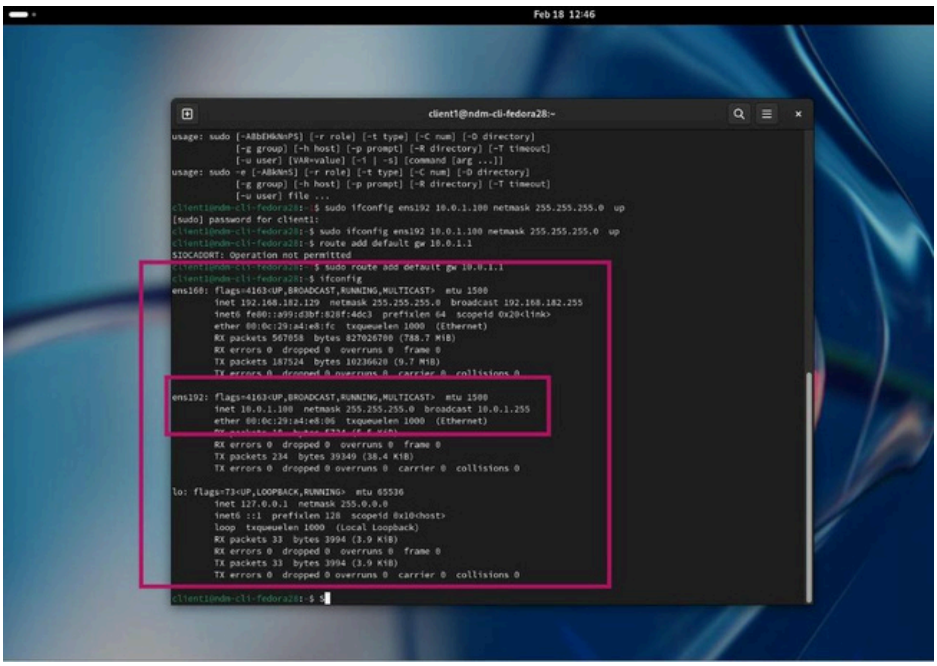
4. A default gateway can be assigned by using the command;

route add default gw <Gateway IP>



5. Step 4 – Verify using

ifconfig



Editing Network Interface Configuration File

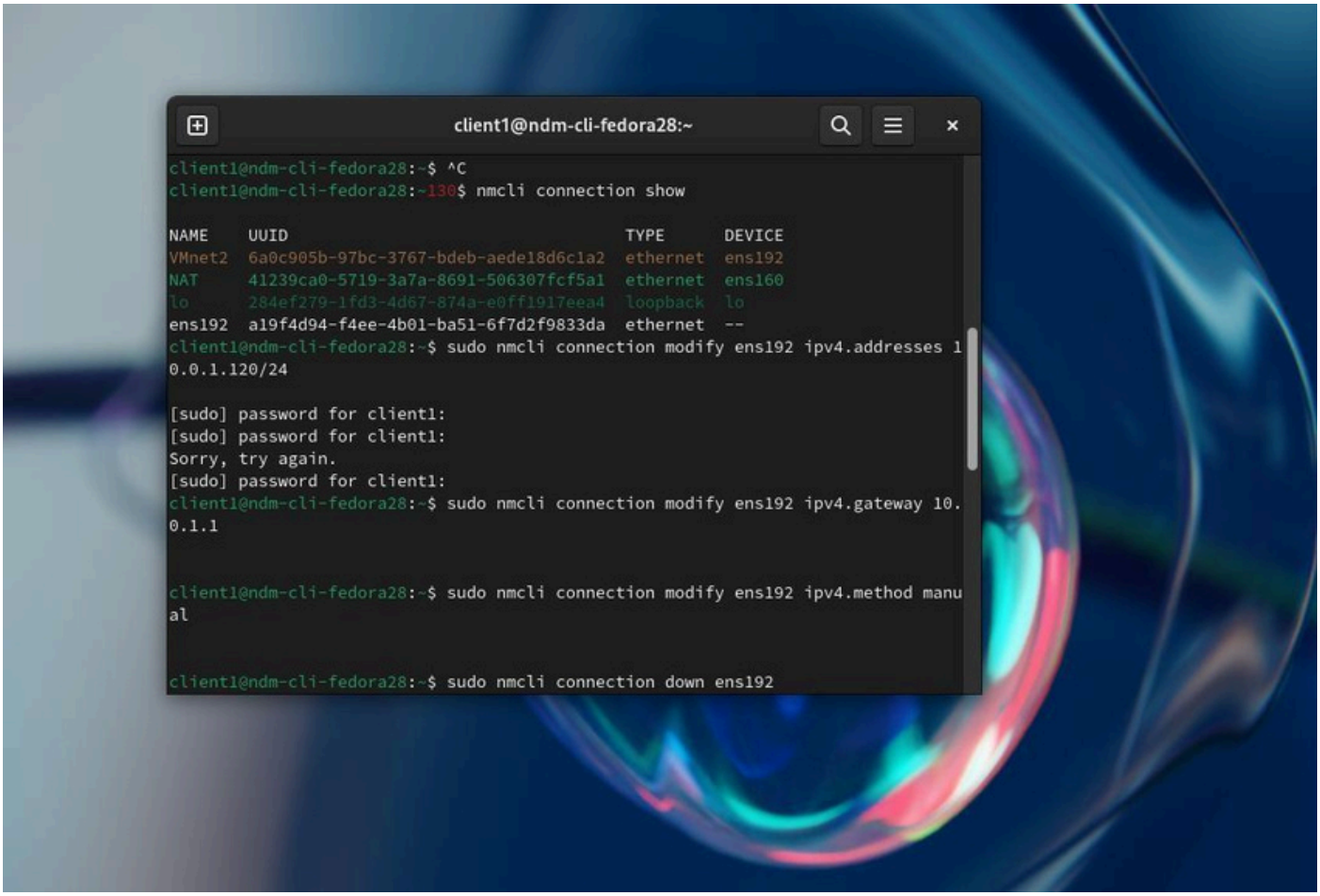
Go to Network Scripts Directory

cd /etc/sysconfig/network-scripts/

ls

Then

nmcli connection show



Step 1 – Set IP Address

sudo nmcli connection modify ens192 ipv4.addresses 10.0.1.120/24

- **Step 2 – Set Gateway**

sudo nmcli connection modify ens192 ipv4.gateway 10.0.1.1

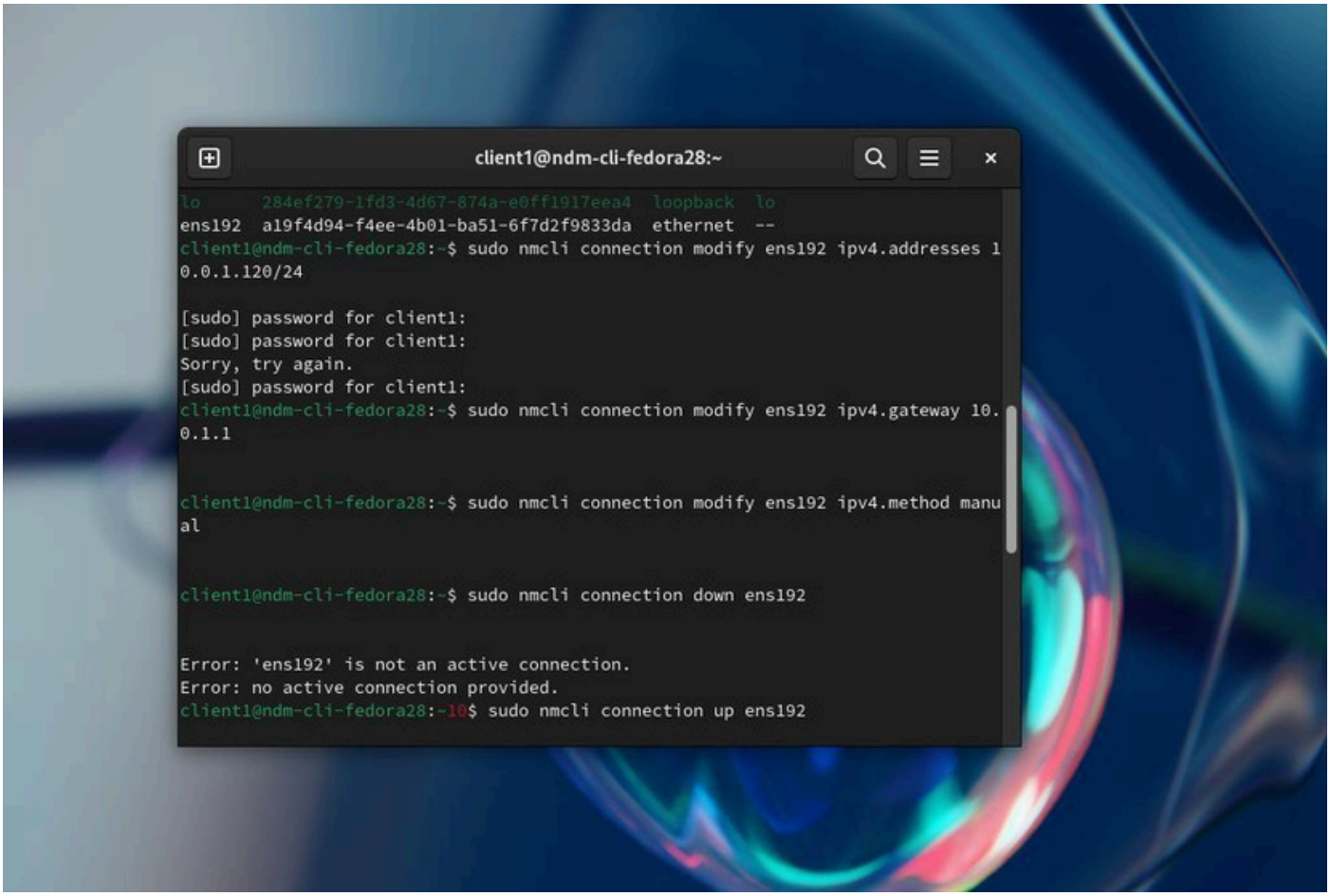
- **Step 3 – Set Method to Manual**

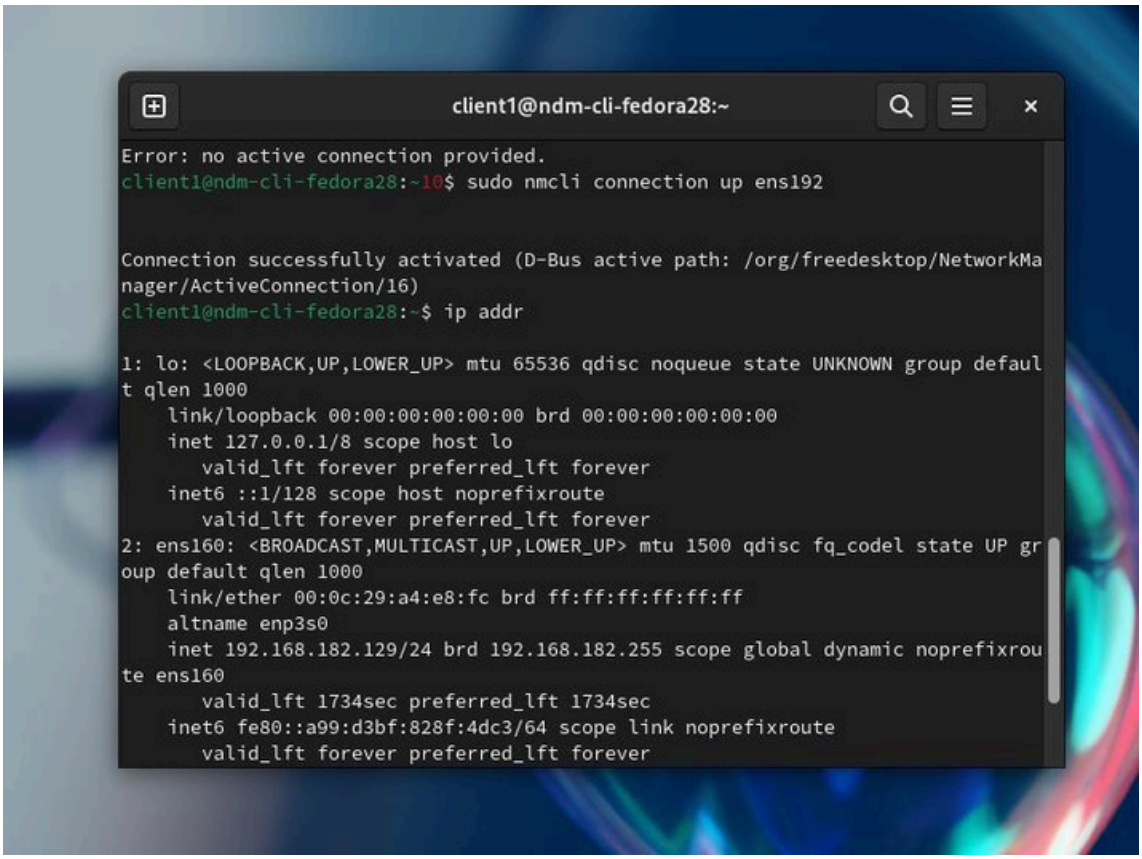
sudo nmcli connection modify ens192 ipv4.method manual

- **Step 4 – Restart Connection**

sudo nmcli connection down ens192

sudo nmcli connection up ens192



Step 5 – Verify

ip addr